

Regular functions are v -adically analytic

The formal expansion of $f \in \mathcal{O}_{X,P}$ in local coordinates converges on a v -adic neighbourhood of P — proved from one fixed-point lemma, applied twice

with the formal group of an elliptic curve as the worked case

1. Statement

Throughout, K is a number field, v a place of K , K_v the completion, and $|\cdot|$ an absolute value on K_v inducing its topology. **No distinction is made between the archimedean and the non-archimedean case:** the only properties used are that $|\cdot|$ is multiplicative, satisfies the triangle inequality, and that K_v is complete.

Standing hypotheses. X is a variety over K (a separated scheme of finite type), and $P \in X(K)$ is a **regular** point of X , of local dimension $n = \dim_P X$. Since K is perfect, regular is the same as smooth. Regularity is not a convenience but a necessity: it is exactly what makes “local coordinates $x_1, \dots, x_n$ at P ” exist, and Section 7 says what happens without it.

Local coordinates at P means a regular system of parameters: elements $x_1, \dots, x_n \in \mathfrak{m}_P \subset \mathcal{O}_{X,P}$ whose classes form a K -basis of $\mathfrak{m}_P/\mathfrak{m}_P^2$.

Because $\mathcal{O}_{X,P}$ is a regular local ring of dimension n containing its residue field K , Cohen’s structure theorem gives a canonical isomorphism

$$\hat{\mathcal{O}}_{X,P} \cong K[[x_1, \dots, x_n]]$$

carrying x_i to x_i , and $\mathcal{O}_{X,P} \hookrightarrow \hat{\mathcal{O}}_{X,P}$ is injective by Krull’s intersection theorem. The **formal expansion** of $f \in \mathcal{O}_{X,P}$ is its image

$$F = \sum_{\alpha} a_{\alpha} x^{\alpha} \in K[[x_1, \dots, x_n]], \quad a_{\alpha} \in K.$$

The set $X(K_v)$ carries the v -adic topology, defined through any affine chart and independent of the choice. For $\rho > 0$ write $D(\rho) = \{a \in K_v^n : |a_i| < \rho \text{ for all } i\}$.

Theorem. With the hypotheses above, there exist $\rho > 0$ and a v -adically open $\Omega \subseteq X(K_v)$ with $P \in \Omega$ such that

- (a) $\sum_{\alpha} |a_{\alpha}| \rho^{|\alpha|} < \infty$; in particular F converges absolutely at every point of $D(\rho)$;
- (b) $x = (x_1, \dots, x_n)$ is defined on Ω and maps it homeomorphically onto an open subset of $D(\rho)$ containing 0;
- (c) f is defined at every $Q \in \Omega$ and

$$f(Q) = \sum_{\alpha} a_{\alpha} x(Q)^{\alpha},$$

the series converging absolutely.

The conclusion holds for **every** choice of local coordinates at P , and ρ and Ω may be shrunk freely.

The proof occupies Section 2 through Section 6. The engine is a single fixed-point lemma (Section 3), applied twice: once in a Banach algebra of power series, where it produces the expansion and its conver-

gence, and once in K_v itself, where it produces the neighbourhood Ω . That the two applications agree is a one-line functoriality argument, and it is what glues “formal” to “analytic”.

2. Convergent power series

For $\rho > 0$ and $F = \sum_{\alpha} a_{\alpha} x^{\alpha} \in K_v[[x_1, \dots, x_n]]$ set

$$\|F\|_{\rho} = \sum_{\alpha} |a_{\alpha}| \rho^{|\alpha|} \in [0, \infty], \quad K_v\langle \rho \rangle = \{F : \|F\|_{\rho} < \infty\},$$

and let $K_v\{x\} = \bigcup_{\rho>0} K_v\langle \rho \rangle$ be the ring of **convergent** power series. (In the non-archimedean case the more usual condition is $|a_{\alpha}| \rho^{|\alpha|} \rightarrow 0$; since the number of α with $|\alpha| = d$ grows polynomially in d , the two conditions define the same ring $K_v\{x\}$, differing only by a shrinking of ρ .)

Lemma 2.1. $(K_v\langle \rho \rangle, \|\cdot\|_{\rho})$ is a commutative Banach K_v -algebra: the norm satisfies $\|\lambda F\|_{\rho} = |\lambda| \|F\|_{\rho}$, $\|F + G\|_{\rho} \leq \|F\|_{\rho} + \|G\|_{\rho}$, $\|FG\|_{\rho} \leq \|F\|_{\rho} \|G\|_{\rho}$, and $K_v\langle \rho \rangle$ is complete. For $a \in D(\rho)$ the evaluation $\text{ev}_a : K_v\langle \rho \rangle \rightarrow K_v$ is a K_v -algebra homomorphism with $|\text{ev}_a(F)| \leq \|F\|_{\rho}$.

Proof. Submultiplicativity: the γ -th coefficient of FG is $\sum_{\alpha+\beta=\gamma} a_{\alpha} b_{\beta}$, so $\|FG\|_{\rho} \leq \sum_{\gamma} \sum_{\alpha+\beta=\gamma} |a_{\alpha}| |b_{\beta}| \rho^{|\gamma|} = \|F\|_{\rho} \|G\|_{\rho}$. Completeness: $K_v\langle \rho \rangle$ is the space of coefficient families that are absolutely summable against the weights $\rho^{|\alpha|}$, and K_v is complete. Evaluation: $|\sum a_{\alpha} a^{\alpha}| \leq \sum |a_{\alpha}| \rho^{|\alpha|}$, and absolute convergence licenses rearranging the product. ■

Lemma 2.2 (shrinking). If $F(0) = 0$ and $\|F\|_{\rho_0} < \infty$, then for $0 < \rho \leq \rho_0$,

$$\|F\|_{\rho} \leq (\rho/\rho_0) \|F\|_{\rho_0}.$$

In particular $\|F\|_{\rho} \rightarrow 0$ as $\rho \rightarrow 0$.

Proof. Every α occurring has $|\alpha| \geq 1$, so $\rho^{|\alpha|} = \rho_0^{|\alpha|} (\rho/\rho_0)^{|\alpha|} \leq \rho_0^{|\alpha|} (\rho/\rho_0)$. ■

Lemma 2.3 (units). If $F \in K_v\{x\}$ and $F(0) \neq 0$ then F is a unit of $K_v\{x\}$. Hence $K_v\{x\}$ is a local ring with maximal ideal $\{F : F(0) = 0\}$.

Proof. Rescale so $F(0) = 1$ and write $F = 1 - G$ with $G(0) = 0$. By Lemma 2.2 choose ρ with $\|G\|_{\rho} \leq 1/2$. Then $\sum_{k \geq 0} G^k$ converges in the Banach algebra $K_v\langle \rho \rangle$ and is inverse to F . ■

Lemma 2.4 (composition). Let $F \in K_v\{y_1, \dots, y_k\}$ and $G_1, \dots, G_k \in K_v\{x\}$ with $G_i(0) = 0$. Then the formal composite $F(G_1, \dots, G_k)$ lies in $K_v\{x\}$.

Proof. Pick σ with $\|F\|_{\sigma} < \infty$, then by Lemma 2.2 pick ρ with $\|G_i\|_{\rho} \leq \sigma$ for all i . Then $\|F(G)\|_{\rho} \leq \sum_{\beta} |b_{\beta}| \prod_i \|G_i\|_{\rho}^{\beta_i} \leq \sum_{\beta} |b_{\beta}| \sigma^{|\beta|} = \|F\|_{\sigma} < \infty$; absolute convergence justifies the rearrangement. ■

3. The fixed-point lemma

This is the implicit function theorem, stated so that it can be applied in any Banach algebra at once. Fix $m = n + r$ and convergent power series $g_1, \dots, g_r \in K_v\{T_1, \dots, T_m\}$ with

$$g_i(0) = 0 \quad \text{and} \quad M := \left(\frac{\partial g_i}{\partial T_{n+j}}(0) \right)_{1 \leq i, j \leq r} \in \text{GL}_r(K_v).$$

Write $T = (T', T'')$ with $T' = (T_1, \dots, T_n)$ and $T'' = (T_{n+1}, \dots, T_m)$, and expand

$$g(T', T'') = AT' + MT'' + Q(T', T''),$$

where $A \in M_{r \times n}(K_v)$ collects the remaining linear terms and Q collects all terms of total degree ≥ 2 .

Lemma 3.1. There are $\varepsilon, \delta > 0$, depending only on g , with the following property. Let $(B, \|\cdot\|)$ be any commutative Banach K_v -algebra whose norm satisfies $\|\lambda b\| = |\lambda| \|b\|$ and $\|bb'\| \leq \|b\| \|b'\|$. Then for every $u \in B^n$ with $\|u_i\| \leq \varepsilon$ there is a **unique** $t \in B^r$ with

$$\|t_j\| \leq \delta \quad \text{and} \quad g(u, t) = 0 \text{ in } B^r.$$

Proof. Let $R > 0$ be less than the radius of convergence of all the g_i , so that $C_0 := \sum_{|\beta| + |\gamma| \geq 2} |q_{\beta\gamma}| R^{|\beta| + |\gamma| - 2}$ and $C_1 := \sum_{|\beta| + |\gamma| \geq 2} |\gamma| |q_{\beta\gamma}| R^{|\beta| + |\gamma| - 2}$ are both finite, where $q_{\beta\gamma}$ are the coefficients of Q . (If the g_i are polynomials, as they will be in Section 6, these are finite sums and there is nothing to check.) Put

$$c = \max_i \sum_j |(M^{-1})_{ij}|, \quad a = \max_i \sum_j |A_{ij}|,$$

so that $\|M^{-1}b\| \leq c \max_j \|b_j\|$ and $\|Ab\| \leq a \max_j \|b_j\|$ in any B : these bounds use only the triangle inequality and $\|\lambda b\| = |\lambda| \|b\|$, so they are **uniform in B** .

Choose

$$\delta \leq \min\left(R, \frac{1}{4cC_0}, \frac{1}{2cC_1}\right), \quad \varepsilon \leq \min\left(\delta, \frac{\delta}{2ca}\right).$$

Let $u \in B^n$ with $\|u_i\| \leq \varepsilon$ and let

$$\mathcal{B} = \{t \in B^r : \|t_j\| \leq \delta \text{ for all } j\},$$

a closed ball in B^r with the max norm, hence a complete metric space. For $t \in \mathcal{B}$ all the series $g_i(u, t)$ converge absolutely in B , since $\|u_i\|, \|t_j\| \leq \delta \leq R$. Define

$$N(t) = t - M^{-1}g(u, t) = -M^{-1}Au - M^{-1}Q(u, t),$$

the second equality by the expansion of g . A fixed point of N is exactly a solution of $g(u, t) = 0$, since M^{-1} is invertible.

N maps $\mathcal{B}$ into itself. With $s = \delta$ (note $\varepsilon \leq \delta$),

$$\|Q(u, t)\| \leq \sum_{|\beta| + |\gamma| \geq 2} |q_{\beta\gamma}| \varepsilon^{|\beta|} \delta^{|\gamma|} \leq s^2 \sum |q_{\beta\gamma}| R^{|\beta| + |\gamma| - 2} = C_0 \delta^2,$$

using $\varepsilon, \delta \leq R$ and $|\beta| + |\gamma| \geq 2$. Hence

$$\|N(t)\| \leq ca\varepsilon + cC_0\delta^2 \leq \frac{\delta}{2} + \frac{\delta}{4} \leq \delta.$$

N is a contraction. For $t, t' \in \mathcal{B}$ only the terms with $|\gamma| \geq 1$ survive in the difference, and telescoping gives $\|t^\gamma - t'^\gamma\| \leq |\gamma| \delta^{|\gamma| - 1} \max_j \|t_j - t'_j\|$. Therefore

$$\|N(t) - N(t')\| \leq c \sum_{|\gamma| \geq 1, |\beta| + |\gamma| \geq 2} |q_{\beta\gamma}| \varepsilon^{|\beta|} |\gamma| \delta^{|\gamma| - 1} \max_j \|t_j - t'_j\| \leq cC_1 \delta \max_j \|t_j - t'_j\| \leq \frac{1}{2} \max_j \|t_j - t'_j\|,$$

where the middle step uses $\varepsilon^{|\beta|} \delta^{|\gamma| - 1} \leq \delta \cdot R^{|\beta| + |\gamma| - 2}$, valid because $|\beta| + |\gamma| - 1 \geq 1$ and $\varepsilon \leq \delta \leq R$.

By the Banach fixed-point theorem N has a unique fixed point in $\mathcal{B}$. ■

Lemma 3.2 (functoriality). Let $\chi : B \rightarrow B'$ be a K_v -algebra homomorphism with $\|\chi(b)\| \leq \|b\|$, and let u, t be as in Lemma 3.1. Then $\chi(t)$ is the unique solution attached to $\chi(u)$.

Proof. χ is continuous, so it commutes with the convergent series defining g ; hence $g(\chi(u), \chi(t)) = \chi(g(u, t)) = 0$, and $\|\chi(t_j)\| \leq \|t_j\| \leq \delta$, and $\|\chi(u_i)\| \leq \varepsilon$. Uniqueness in Lemma 3.1 applied over B' finishes. ■

4. Formal uniqueness

Lemma 4.1. Let L be any field and $g_1, \dots, g_r \in L[[T_1, \dots, T_m]]$ with $g(0) = 0$ and $M \in \text{GL}_r(L)$ as above. Then there is **at most one** $t \in (x)^r \subset L[[x_1, \dots, x_n]]^r$ with $g(x, t) = 0$, where (x) denotes the maximal ideal.

Proof. Suppose t, t' are two solutions and $t \equiv t' \pmod{(x)^d}$ with $d \geq 1$; write $e = t' - t \in (x)^d$. Taylor expansion in the second block of variables gives

$$0 = g(x, t') - g(x, t) = \sum_j \frac{\partial g}{\partial T_{n+j}}(x, t) e_j + O(e^2).$$

Now $(\partial g / \partial T_{n+j})(x, t) \equiv M_{\cdot j} \pmod{(x)}$ because x and t lie in (x) , so the sum is $\equiv Me \pmod{(x)^{d+1}}$; and $O(e^2) \in (x)^{2d} \subseteq (x)^{d+1}$. Hence $Me \equiv 0 \pmod{(x)^{d+1}}$ and, M being invertible, $e \in (x)^{d+1}$, i.e. $t \equiv t' \pmod{(x)^{d+1}}$. Both solutions lie in $(x)^1$, so induction gives $t = t'$. ■

Two consequences used below. First, the solution produced by Lemma 3.1 in $B = K_v\langle \rho \rangle$ is **the** formal solution. Second, if the g_i have coefficients in a subfield $K \subseteq K_v$, then solving degree by degree — which is possible, and determined, by the same computation — produces a solution in $K[[x]]^r$; by uniqueness it coincides with the analytic one. **So the convergent solution automatically has coefficients in K .**

5. The local presentation of a smooth point

Proposition 5.1. There are an affine open $U \subseteq X$ containing P , a closed embedding $U \hookrightarrow \mathbb{A}^m$ sending P to the origin, and polynomials $g_1, \dots, g_r \in I(U) \subset K[T_1, \dots, T_m]$ with $r = m - n$, such that after renumbering the coordinates:

- (i) $M = (\partial g_i / \partial T_{n+j})(0)_{i,j}$ is invertible;
- (ii) $U = V(g_1, \dots, g_r)$ in some Zariski neighbourhood of P ;
- (iii) $T_1|_U, \dots, T_n|_U$ is a system of local coordinates at P .

Proof. Choose any affine open $U = \text{Spec } A$ containing P and a closed embedding $U \subset \mathbb{A}^m$ with $P \mapsto 0$; let $I = I(U)$. Since P is a smooth point of local dimension n , the Jacobian criterion says the matrix $(\partial g / \partial T_j(0))_{g \in I}$ has rank $m - n = r$. Pick $g_1, \dots, g_r \in I$ realising that rank and renumber the T_j so that (i) holds.

For (ii): $V := V(g_1, \dots, g_r)$ contains U near P and is cut out by r equations whose differentials at 0 are independent, so $\mathcal{O}_{V,P}$ is regular of dimension $m - r = n$; in particular it is a domain. The surjection $\mathcal{O}_{V,P} \twoheadrightarrow \mathcal{O}_{U,P}$ has kernel a prime $\mathfrak{p}$ with $\dim \mathcal{O}_{V,P}/\mathfrak{p} = n = \dim \mathcal{O}_{V,P}$, so $\text{ht } \mathfrak{p} = 0$ and $\mathfrak{p} = 0$. Thus $\mathcal{O}_{V,P} \cong \mathcal{O}_{U,P}$, and two finite-type schemes with the same local ring at P agree on a neighbourhood.

For (iii): the relations $dg_i = \sum_j (\partial g_i / \partial T_j)(0) dT_j = 0$ hold in $\mathfrak{m}_P / \mathfrak{m}_P^2$, and by (i) they express $dT_{n+1}, \dots, dT_m$ in terms of $dT_1, \dots, dT_n$. Hence $dT_1, \dots, dT_n$ span the n -dimensional space $\mathfrak{m}_P / \mathfrak{m}_P^2$, so they are a basis. ■

6. Proof of the Theorem

Step 0: reduction to K_v . Base change to K_v commutes with everything in sight: P remains a smooth K_v -point of X_{K_v} , the local coordinates remain local coordinates, and the formal expansion of f over K_v is the image of the one over K — **the same coefficients**. So we may compute over K_v , and it is only the convergence statement that is at issue.

Step 1: the special coordinates. Take $U, g_i, m = n + r$ as in Proposition 5.1, and write $x_i^0 = T_i|_U$ for $i \leq n$, which by 5.1(iii) are local coordinates at P . Apply **Lemma 3.1 with $B = K_v\langle\rho\rangle$** (the power series algebra in n variables) and

$$u = (x_1, \dots, x_n), \quad \|x_i\|_\rho = \rho \leq \varepsilon.$$

This yields $\varphi = (\varphi_1, \dots, \varphi_r) \in K_v\langle\rho\rangle^r$ with $\|\varphi_j\|_\rho \leq \delta$ and

$$g_i(x_1, \dots, x_n, \varphi_1, \dots, \varphi_r) = 0 \quad (i = 1, \dots, r).$$

Applying Lemma 3.2 to $\text{ev}_0 : K_v\langle\rho\rangle \rightarrow K_v$ shows $\varphi(0)$ is the unique small solution of $g(0, t) = 0$, which is $t = 0$; hence $\varphi_j(0) = 0$. By Lemma 4.1 the φ_j are **the** formal solutions, so by 5.1(ii) they are the formal expansions of $T_{n+j}|_U$, and by Section 4 they have coefficients in K .

Step 2: the neighbourhood. Apply **Lemma 3.1 with $B = K_v$** itself. For each $a \in D(\varepsilon)$ there is a unique $t(a) \in K_v^r$ with $|t_j(a)| \leq \delta$ and $g(a, t(a)) = 0$; and by Lemma 3.2 applied to ev_a — which is norm-decreasing by Lemma 2.1 — we get $t(a) = \varphi(a)$. Set

$$\Omega = \{Q \in U(K_v) : |T_i(Q)| < \varepsilon \quad (i \leq n), \quad |T_{n+j}(Q)| < \delta \quad (j \leq r)\}.$$

Then Ω is open in $X(K_v)$ (it is $U(K_v)$, open, intersected with an open box), contains P , and by the previous sentence

$$\Omega = \{(a, \varphi(a)) : a \in D(\varepsilon)\}.$$

So $x = (T_1, \dots, T_n)$ maps Ω bijectively onto $D(\varepsilon)$, continuously in both directions — the inverse being $a \mapsto (a, \varphi(a))$, continuous because a convergent power series is continuous on its polydisc of convergence. This is (b), with $\rho = \varepsilon$.

Step 3: every regular function. Let

$$\mathcal{E} : \mathcal{O}_{X,P} \longrightarrow K_v[[x_1, \dots, x_n]]$$

be the expansion map, a K_v -algebra homomorphism. The affine coordinate ring A of U is generated by the $T_i|_U$, whose expansions are $x_1, \dots, x_n$ and $\varphi_1, \dots, \varphi_r$, all in $K_v\langle\rho\rangle$. As $K_v\langle\rho\rangle$ is a K_v -algebra, $\mathcal{E}(A) \subseteq K_v\langle\rho\rangle$. A general $f \in \mathcal{O}_{X,P}$ is g/h with $g, h \in A$ and $h(P) \neq 0$; then $\mathcal{E}(h)$ is convergent with constant term $h(P) \neq 0$, hence invertible in $K_v\{x\}$ by Lemma 2.3, so

$$F = \mathcal{E}(f) = \mathcal{E}(g) \cdot \mathcal{E}(h)^{-1} \in K_v\{x\}.$$

That is (a), after shrinking ρ .

Step 4: the identity on Ω . Shrink Ω so that $|h(Q) - h(P)| < |h(P)|$ on it, which is possible since h is continuous and $h(P) \neq 0$; then h does not vanish on Ω and f is defined there. Fix $Q \in \Omega$ and let $a = x(Q) \in D(\rho)$. Under the norm-decreasing homomorphism ev_a we have $x_i \mapsto T_i(Q)$ and, by Step 2, $\varphi_j \mapsto \varphi_j(a) = T_{n+j}(Q)$. Hence for every $p \in A$, $\text{ev}_a(\mathcal{E}(p)) = p(Q)$, both sides being the same polynomial in the coordinates of Q . Applying this to g and h and using that ev_a is a ring homomorphism,

$$F(a) = \text{ev}_a(\mathcal{E}(g)) \cdot \text{ev}_a(\mathcal{E}(h))^{-1} = g(Q)/h(Q) = f(Q).$$

That is (c).

Step 5: arbitrary local coordinates. Let $x'_1, \dots, x'_n$ be any system of local coordinates at P . Each x'_i lies in $\mathcal{O}_{X,P}$, so by Steps 3–4 its expansion G_i in the coordinates x^0 is convergent, and $G_i(0) = 0$ because

$x'_i \in \mathfrak{m}_P$. The matrix $J = (\partial G_i / \partial x_j(0))$ is the matrix of the change of basis of $\mathfrak{m}_P / \mathfrak{m}_P^2$ from x^0 to x' , hence invertible. Apply Lemma 3.1 to the convergent system

$$\tilde{g}_i(y, t) = G_i(t) - y_i \quad (i = 1, \dots, n),$$

for which $\partial \tilde{g}_i / \partial t_j(0) = J$ is invertible: it produces a convergent H with $H(0) = 0$ and $G(H(y)) = y$, and Lemma 4.1 gives $H(G(x)) = x$ as well. Since the identification $K[[x']] \cong \hat{\mathcal{O}}_{X,P} \cong K[[x^0]]$ sends x'_i to G_i , the expansion of f in the coordinates x' is

$$F' = F \circ H,$$

which is convergent by Lemma 2.4. Statements (a)–(c) for x' follow, after replacing Ω by $\Omega \cap x'^{-1}(D(\rho'))$. ■

7. Complements

7.1. The expansion map, and a rigidity corollary

Corollary 7.1. $\mathcal{E} : \mathcal{O}_{X,P} \rightarrow K_v\{x\}$ is an injective local homomorphism of K -algebras. Consequently, if $f \in \mathcal{O}_{X,P}$ vanishes on some v -adic neighbourhood of P in $X(K_v)$, then $f = 0$ in $\mathcal{O}_{X,P}$, hence f vanishes on the irreducible component of X through P .

Proof. Injectivity is Krull. If f vanishes near P , then by (c) the convergent series F vanishes at every point of a polydisc of positive radius; a convergent power series with that property has all coefficients zero — for archimedean v this is the identity theorem, for non-archimedean v it follows from Strassmann’s theorem by induction on n . So $F = 0$ and $f = 0$. ■

This is the statement that gets used whenever one argues that an algebraic identity may be **checked v -adically on a small neighbourhood**: it is legitimate exactly because the expansion map is injective.

7.2. $X^{\text{sm}}(K_v)$ is a K_v -analytic manifold

Steps 2 and 5 say more than the theorem: the charts $x : \Omega \rightarrow D(\rho)$ obtained at the various smooth rational points are analytically compatible, since the transition maps are the H of Step 5, convergent with convergent inverses. Hence $X^{\text{sm}}(K_v)$ is a K_v -analytic manifold of dimension n , and every regular function on X is analytic on it. For archimedean v this is the classical statement that a smooth variety over $\mathbb{R}$ or $\mathbb{C}$ is a real or complex manifold; for non-archimedean v it is the statement underlying every “ p -adic density” argument.

7.3. The integral refinement, for almost all v

In practice one wants more than **some** radius: one wants the whole residue disc, and coefficients in $\mathcal{O}_v$. This holds for all but finitely many v , and the proof is the same contraction run in a different complete ring.

Proposition 7.2. Suppose v is non-archimedean, $g_1, \dots, g_r \in \mathcal{O}_v[T_1, \dots, T_m]$ and $M \in \text{GL}_r(\mathcal{O}_v)$. Then the solution φ of Step 1 lies in $\mathcal{O}_v[[x_1, \dots, x_n]]^r$, and the parametrisation $a \mapsto (a, \varphi(a))$ identifies $\mathfrak{m}_v^n$ with the residue disc of P in $X(K_v)$.

Proof. Run the map $N(t) = -M^{-1}Ax - M^{-1}Q(x, t)$ of Section 3 on $J \cdot \mathcal{O}_v[[x]]^r$, where $J = (x_1, \dots, x_n)$. All coefficients stay in $\mathcal{O}_v$ because M^{-1}, A, Q have entries there. If $t \equiv t' \pmod{J^d}$ then each surviving term $q_{\beta\gamma}x^\beta(t^\gamma - t'^\gamma)$ lies in $J^{|\beta| + |\gamma| - 1 + d} \subseteq J^{d+1}$, since $|\beta| + |\gamma| \geq 2$; so N is a contraction for the J -adic metric, which is complete. Its fixed point is φ , by Lemma 4.1. ■

Given a model of X over $\mathcal{O}_K[1/N]$ that is smooth along P , the hypotheses of 7.2 hold for every $v \nmid N$. This — not the qualitative theorem — is the form used when one needs uniformity in v .

7.4. What regularity was for

If P is a singular point there is no system of n local coordinates, and the statement as posed does not typecheck: $\hat{\mathcal{O}}_{X,P}$ is not a power series ring. For the nodal cubic $y^2 = x^2(x+1)$ at the origin one has $\hat{\mathcal{O}}_{X,P} \cong K[[u, w]]/(uw)$, and $X(K_v)$ near P is two analytic branches crossing — not a manifold. What survives is the statement **branch by branch**: on the normalisation, where the preimages of P are smooth, the theorem applies unchanged. Similarly, if P is a closed point whose residue field is a finite extension L/K rather than K , one replaces v by the places $w \mid v$ of L and the theorem holds over each L_w .

Nothing in Section 2 through Section 6 used that K is a number field beyond the completeness of K_v . **The theorem holds verbatim over any field complete with respect to an absolute value** — $\mathbb{C}_p$, a Laurent series field, a completion of a function field.

8. Worked example: the formal group of an elliptic curve

Let $E : y^2 = x^3 + ax + b$ over K , and let $P = \mathcal{O}$ be the point at infinity, a smooth rational point with $n = 1$. In the standard affine chart at infinity put

$$z = -x/y, \quad w = -1/y,$$

so that $x = z/w$ and $y = -1/w$; both z and w are regular at $\mathcal{O}$, and z is a uniformiser, hence a local coordinate. Substituting into the Weierstrass equation and clearing denominators gives the exact relation

$$w = z^3 + azw^2 + bw^3,$$

that is $g(z, w) = 0$ for $g = T_2 - T_1^3 - aT_1T_2^2 - bT_2^3$, with $g(0, 0) = 0$ and $\partial g / \partial T_2(0, 0) = 1$, invertible. **This is precisely the situation of Lemma 3.1 with $n = r = 1$** , so the theorem applies: the formal expansion $w(z) \in K[[z]]$ converges on a v -adic disc around $z = 0$, and there parametrises $E(K_v)$ near $\mathcal{O}$.

For $E : y^2 = x^3 - 16x + 16$ (a model of 37a1), iterating the relation gives

$$w(z) = z^3 - 16z^7 + 16z^9 + 512z^{11} - 1280z^{13} - 19712z^{15} + \dots,$$

the coefficients lying in $\mathbb{Z}[a, b]$ as Proposition 7.2 predicts. Taking $v = 5$ and the point $Q = 8P_0$, where $P_0 = (0, 4)$ generates $E(\mathbb{Q})$, one has $v_5(z(Q)) = 1$ and $v_5(w(Q)) = 3$; truncating the series at order N and evaluating at $z(Q)$ gives an error of valuation

truncation order N	10	20	30	40
$v_5(w(Q) - w_N(z(Q)))$	14	23	33	43

— growing linearly in N , which is exactly what convergence at a point of valuation 1 predicts. The computation is in `analytic-local.gp`.

Why this example is the one that matters here. The first proof of Lemma B' in `selmer-involution.typ` opens with: “the formal group gives a finite-index subgroup of $E(K)$ isomorphic to $(\mathfrak{m}_K, +) \cong \mathbb{Z}_p^{[K:\mathbb{Q}_p]}$ ”. That sentence **is** the theorem of Section 1, specialised to $X = E$, $P = \mathcal{O}$, $f = w$: what makes the formal group a subgroup of $E(K_v)$ rather than a formal object is precisely that $w(z)$, and the formal group law, converge. The count $\dim E(\mathbb{Q}_v)/2 = \dim M^{G_v} + [\mathbb{Q}_v : \mathbb{Q}_2]$ rests on it.

9. Where this is used elsewhere in these notes

The theorem is the licence for three moves that are made constantly and rarely stated.

move	what licenses it
treating $X(\mathbb{Q}_p)$ as a p -adic manifold and speaking of density of a subset	Section 7.2
parametrising a residue disc by $\mathbb{Z}_p^n$ and pushing a measure around	Proposition 7.2
checking an algebraic identity on a p -adic neighbourhood and concluding it globally	Corollary 7.1

The first two are what `kummer-padic-density.typ` and `kummer2.gp` assume throughout: `densegroup` asks whether a subgroup of $E(\mathbb{Q}_p)$ is **dense**, a question that presupposes the analytic structure of Section 7.2 and is answered by reducing to the formal group, i.e. to Section 8. The third is what allows a local computation to prove a global statement.